Aadarsh Ramachandiran

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Education

Examination	University	Institute	Year	CPI / %
Bachelors	IIT Madras	IIT Madras	2027	9.39
Intermediate	CBSE	Maharishi International Residential School	2022	97.60%
Matriculation	CBSE	Maharishi International Residential School	2020	95.80%

Scholastic Achievements

- Selected for **NUS Young Fellowship**, a prestigious academic immersion program for top undergraduate scholars *2025*
- Achieved **All India Rank 849** in **Joint Entrance Examination-Advanced** among 180,000+ candidates. *2023*
- Secured **All India Rank 269** in the **Joint Entrance Examination-Mains** among 950,000+ candidates. *2023*
- Ranked in the **Top 1%** nationally out of 50,000+ students in **National Standard Examination in Physics**. *2022*
- Attained **Top 1%** in the state in **National Standard Examination in Astronomy**. *2022*
- Awarded **Kishore Vaigyanik Protsahan Yojana(KVPY) Fellowship** (**All India Rank 304** among 50k+) *2021*
- Awarded **National Talent Search Examination Fellowship** by NCERT, Government of India. *2020*

Research Experience

Language-based User-Centric Robot Trajectory Adaptation *Summer 2025*
Indian Institute of Science *Guide: Prof. Ravi Prakash*

- Designed a framework integrating **Large Language Models (LLMs)** with **diffusion-based motion planners** for natural language-conditioned trajectory adaptation in robotic manipulators.
- Implemented **zero-shot, test-time adaptation** for constraint satisfaction and fine-grained user control, enabling generalization to novel tasks and environments.
- Performed critical analysis of prior works and demonstrated improvements in **interpretability** and **user-centric guidance** via cost- and language-driven control.

Key Projects

Le X-O *Repository*
LeRobot Worldwide Hackathon *Summer 2025*

- Developed **Le X-O**, a robotic arm agent capable of playing a full game of Tic-Tac-Toe against humans, integrating **reasoning, perception, and control** in real time.
- Designed a **hierarchical architecture** combining a high-level **LLM planner** with a low-level **Action Chunking Transformer (ACT)** policy for precise execution.
- Enhanced ACT with **task-based language-conditioning**, enabling **multi-task generalization** from minimal demonstrations.
- Demonstrated a fully working prototype and interpreted the influence of vision and joint states on action decisions.

Agnirath
Race Strategist Engineer *2024*

- Developed race strategy for the **3,000 km World Solar Challenge 2025**, maximizing energy efficiency and race outcomes.
- Achieved **10× faster simulations** through **NumPy vectorization**, caching, and multiprocessing for rapid strategy iteration.
- Built a **COBYLA-based constrained optimizer** to generate feasible velocity trajectories under complex race constraints.
- Designed an **MPC controller** to adjust speed in real time using telemetry.
- Reduced battery SoC prediction error by **15%** through field data analysis and nonlinear discharge curve modeling.
- Informed hardware choices (panel tilt, drag area, steering) via simulations, reducing projected energy loss by **8%**.
- Integrated strategy stack with telemetry to ensure reliable communication and optimization during race deployment.

S.A.M.V.I.D
Interface Module —IBot Robotics Club *Fall 2024*

- Designed a **multilingual RAG-powered AI chatbot** and intuitive **UI (HTML, CSS, JS)** for a semi-autonomous museum tour guide robot.
- Improved chatbot's **factual accuracy by 5%** through a **self-corrective RAG retrieval mechanism**.
- Integrated natural language interface with navigation system, enabling **user-driven exhibit selection and guidance**.
- Successfully deployed at **India's first Constitution Museum**, providing interactive guidance to **200+ daily visitors**.

Applied Programming Lab *Repository*
Course Project - Indian Institute of Technology Madras *Spring 2024*

- Implemented a **single-frequency steady-state SPICE-like circuit simulator** in Python using **Modified Nodal Analysis**.
- Optimized keyboard layout via **simulated annealing**, minimizing finger travel distance based on key usage frequency.
- Designed and implemented a **sound source localizer** using the **Delay-and-Sum algorithm** on a simulated microphone array.
- Optimized and benchmarked **trapezoidal integration** with **Cython** and **Python** against NumPy baselines.

Other Projects

ConsumerWise - AI-Powered Nutritional Analysis Tool *Repository*

Self-Project

- Developed **ConsumerWise**, an **AI-driven web application** that provides personalized nutritional insights from **scanned packaged product nutrition labels** and **NutriBot**, a context-aware chatbot to improve Consumer Awareness for **Google Gen-AI Hackathon, 2024**.
- Developed an **end-to-end pipeline** for processing food label images, utilizing OCR to extract data, generating visual graphs and text summaries, and enabling chatbot interactions for comprehensive nutritional insights.
- Deployed the app on **Google Cloud**

Positions of Responsibility

Strategist

IBot Club, IIT Madras *2025*

- Managing club-wide **projects, workshops, competitions, and events** by coordinating efforts across multiple teams.
- Introduced a **robotics and AI community** to discuss latest research and technology, which has grown to 100+ members.
- Leading a team of 8 in **Project GRASP**, exploring Vision Language Action models for **efficient and interpretable general-purpose robotic manipulation**.

Technical Skills

- **Languages:**Python, C, C++ , MATLAB, Verilog, Assembly, HTML, CSS, JS, Bash
- **Tools:**Pytorch, Tensorflow, ROS, MuJoCo, NumPy, Matplotlib, OpenCV, Git, Anaconda

Relevant Coursework

- **AI/ML:** Machine Learning Specialization* (Coursera), Reinforcement Learning
- **Robotics & Control:** Mechanics of Robots, Control Engineering
- **Systems & Signals:** Digital Signal Processing, Microprocessors (Theory & Lab), Applied Programming Lab

** Completed on Coursera*

Extracurricular Activities

- **1st place** out of 80+ participants in **24-hour Multimodal Quantitative Finance** hackathon by GeeksforGeeks on multi-modal time series regression for cryptocurrency price.
- **2nd place** among 70 in **ai to AI** challenge on solving fundamental math problems in Machine learning concepts.
- **3rd place** in a 3-hour hackathon on **multi-label classification of medical specializations from doctor transcripts.**
- **2nd place** among 20+ teams in **24-hour AI hackathon** on **multi-label movie genre classification from plot summaries.**
- **Finalist** out of **50+ teams** in **Product Construct '24**, pitched an innovative solution to **Honda Pvt. Ltd.** for enhancing **Digital Driving License Test Systems** addressing scalability and maintenance costs while ensuring optimal efficiency.